Acoustic fault detection in electrical machines

"For ideathon im doing a project on finding faults in electrical machines using acoustic camera. so i want a research paper which deals with frequencies of electrical machines, the fault frequencies and the normal functioning frequency. I want the most related research paper." *Generated by LeapSpace*

Acoustic-Camera-Based Fault Diagnosis in Electrical Machines at a Glance

Key Findings Table

Topic	Main finding	Machine/fault scope	Most useful citations
Best match for your project	The closest paper in the retrieved set is an experimental induction-motor acoustic study that tested healthy and faulty motors with broken rotor bars, bearing defects, and eccentricities in a reverberant noisy room using five microphones, 44.1 kHz sampling, and about 30 s steady-state recordings; this is the nearest match to an acoustic-camera or microphone-array fault-diagnosis setup [1].	Induction motors; healthy vs multiple faults; realistic noisy environment	[1]
What “normal frequency” means	There is not one universal normal frequency for electrical machines; healthy spectra depend on line frequency, twice-line frequency, line harmonics, slip-related components, load, speed, voltage distortion, resonance, and housing dynamics [2] [3] [4] [5] [6].	Healthy machine baselines	[2] [3] [4] [5] [6]

Topic	Main finding	Machine/fault scope	Most useful citations
Physical origin of healthy sound peaks	Healthy acoustic peaks are explained by electromagnetic excitation such as flux-density harmonics and magnetomotive-force frequencies, often amplified when these excite structural resonances of the stator or housing [7] [8] [9] [10] [11].	Normal operation noise generation	[7] [8] [9] [10] [11]
Fault frequencies as diagnostic markers	Fault diagnosis across sound, vibration, and current mainly relies on harmonics, sidebands, and time-frequency features; faults introduce additional components linked to rotor, stator, bearing, or eccentricity mechanisms [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22].	Bearing, rotor, stator, eccentricity, mechanical faults	[12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22]
Acoustic camera value	Microphone arrays and acoustic-camera-style methods can localize sound sources and help isolate fault-related sound from background noise in noisy industrial settings [23] [24] [25].	Non-contact localization and diagnosis	[23] [24] [25]
Main limitation of literature	The retrieved evidence is much stronger on fault classification accuracy than on reproducible tables of healthy frequencies and exact acoustic fault frequencies; no single paper in the set unifies healthy spectra, all major fault frequencies, and detailed acoustic-camera settings [1] [26] [27] [28] [29] [30] [31] [32].	Evidence gap across the field	[1] [26] [27] [28] [29] [30] [31] [32]

Direct Answer

The **most related paper** in the supplied evidence is the **experimental induction-motor acoustic study** that compared healthy and faulty motors with **broken rotor bars, bearing defects, and eccentricities** in a **reverberant noisy room** using **five microphones**, **44.1 kHz sampling**, and about **30 s steady-state recordings** [1]. It is the closest match to your use case because it combines **real acoustic measurement conditions**, **multiple electrical-machine faults**, and a **microphone-based setup** rather than only simulation or contact sensors [1].

However, if your exact need is **one paper that gives all three together—normal operating frequencies, fault frequencies, and acoustic-camera-style diagnosis**—the retrieved evidence suggests that **no single paper clearly does all of that** [1] [27] [28] [29]. The best practical solution is to use:

1. **The realistic acoustic experiment** as your main project reference [1], and
2. **Companion papers on healthy electromagnetic/acoustic peaks and fault harmonics/sidebands** to build the frequency interpretation layer [2] [3] [7] [8] [9] [18] [19].

So for your ideathon, the strongest evidence-based recommendation is:

- **Primary paper to cite for setup relevance:** the noisy-room multi-microphone induction-motor study [1].
- **Primary papers to cite for normal-frequency understanding:** healthy-operation electromagnetic/acoustic studies explaining peaks from flux harmonics and resonance [7] [8] [9].
- **Primary papers to cite for fault-frequency understanding:** sideband/harmonic studies on broken rotor bars, eccentricity, inter-turn shorts, and mechanical faults [15] [16] [17] [18] [19] [20].

Confidence level: High - The best-match recommendation is directly supported by the retrieved summary for [1], and the limitation that no single fully unified paper was retrieved is explicitly stated in the evidence [1] [27] [28] [29].

Study Scope

This report covers research retrieved on **electrical-machine fault diagnosis using acoustic sensing**, with emphasis on **induction motors and related rotating electrical machines**, and on comparisons between **healthy and faulty spectral behavior** [21] [23]. The evidence spans:

- **Acoustic and microphone-array methods** [23] [24] [33]
- **Healthy-operation acoustic/electromagnetic noise mechanisms** [7] [8] [9] [10] [11]
- **Fault signatures for bearing, rotor-bar, stator, and eccentricity faults** [15] [16] [17] [18] [19] [20]
- **Cross-modal comparison with vibration and current analysis** [34] [35] [36]
- **Feature extraction and classifier pipelines** such as FFT, wavelets, MFCC, PCA/LDA, and neural approaches [30] [37] [38] [39].

Assumptions & Limitations

- The supplied evidence does **not include the exact title/authors** of the most-related paper, so the recommendation is based on the reported experimental description rather than full bibliographic metadata [1].
- The retrieved set does **not provide a single reproducible table** listing healthy acoustic frequencies and all major acoustic fault frequencies in one place [1] [27] [28] [29].
- Many papers emphasize **classification performance** rather than interpretable frequency mapping, so exact band-by-band acoustic fault frequencies are underreported in the retrieved evidence [26] [30] [31] [32].

Suggested Further Research

- Build a **condition-dependent baseline atlas** of healthy motor sound spectra versus load, speed, and supply condition [2] [3] [5].
- Pair **acoustic-camera localization** with **synchronized vibration and current measurements** to validate which acoustic peaks map to which faults [34] [36].
- Publish **explicit frequency tables** for broken rotor bars, bearings, stator faults, and eccentricity under one reproducible acoustic setup [1] [27] [29].
- Pay attention to **low-frequency content below 100 Hz**, which was highlighted as important for condition monitoring [40] [41].
- Focus on **interpretability**, not only classifier accuracy, because this is a clear gap in the current literature [30] [32].

Full Report Body

Introduction and Problem Context

Using an acoustic camera or microphone array to find faults in electrical machines is attractive because it is **non-contact**, can work in **noisy industrial environments**, and can complement vibration and current monitoring rather than replace them [23] [24] [25]. Across the retrieved evidence, acoustic measurements were repeatedly reported as capable of distinguishing **healthy** from **faulty** operation in motors and other rotating machinery [23] [24] [25].

For your project, the central technical issue is **frequency-domain interpretation**. The literature consistently shows that diagnosis is usually based on **spectral peaks, harmonics, sidebands, and time-frequency features** rather than on raw sound alone [12] [13] [14] [21] [22]. In practice, this means your system needs two things:

1. A model of the machine's **healthy baseline spectrum** under known operating conditions [2] [7] [9].
2. A way to detect **new or shifted frequency components** associated with faults such as bearing damage, broken rotor bars, stator defects, or eccentricity [15] [16] [17] [18] [19] [20].

Taken together, the evidence suggests that acoustic-camera-based diagnosis is most useful when it combines **where the sound comes from** with **what frequencies are present** [23] [24] [33].

Confidence level: High - Multiple retrieved sources agree that acoustic sensing is useful and that spectral analysis is the main diagnostic language [21] [22] [23].

Foundational Context: What Frequencies Matter in Electrical-Machine Acoustics

A healthy electrical machine does not emit a single tone. Its acoustic output reflects coupling between **electromagnetic excitation**, **mechanical structure**, and **operating condition** [9] [10] [11]. The retrieved studies describe healthy peaks as arising from **flux-density harmonics**, **magnetomotive-force excitation**, and **structural resonance** of the stator or housing [7] [8] [9].

Across modalities, the most commonly discussed baseline spectral components are:

- **Line frequency**
- **Twice line frequency**
- **Line harmonics**
- **Slip-related components**, including **twice slip frequency** under some conditions [2] [3].

Faults then appear as:

- **Additional harmonics**
- **Sidebands**
- **Envelope-spectrum components**
- **Load-dependent modulations** [15] [16] [18] [19] [36].

This is why “normal functioning frequency” is not a single number. A more accurate concept is a **baseline spectral signature** measured at a known speed, load, and supply condition [4] [5] [6].

An important practical point is that low-frequency sound content can matter: one retrieved acoustic study emphasized that content **below 100 Hz** is essential for condition monitoring [40] [41]. For an acoustic-camera project, this implies that limiting analysis to only high-frequency images may miss useful diagnostic information [40] [41].

Confidence level: High - The underlying mechanism and condition dependence are directly supported across multiple healthy-operation and fault-signature studies [2] [3] [7] [8] [9].

Normal Operating Sound Spectra of Electrical Machines

Healthy-machine acoustic baseline

The healthy baseline in the retrieved studies is dominated by **electromagnetic and harmonic content**, not by a single "machine frequency" [2] [3] [7]. Practical studies report dominant peaks at **line frequency**, **twice line frequency**, and other **line-harmonic components**, with additional dependence on slip and supply condition [2] [3].

Healthy-operation noise studies on induction motors also quantified sound levels and explained spectral peaks as **electromagnetic noise from flux-density harmonics** [7] [8]. These peaks can become especially strong when electromagnetic excitation coincides with **mechanical natural frequencies** of the stator structure [9].

Physical origin of normal acoustic peaks

The studies do not treat acoustic peaks as arbitrary features. Instead, they link them to:

- **Field harmonics**
- **Magnetomotive-force excitation**
- **Torque ripple**
- **Parasitic electromagnetic phenomena**
- **Structural resonance** [5] [10] [11].

This matters for your project because acoustic-camera localization becomes more useful when paired with a physics-based explanation. If a peak comes from electromagnetic forcing exciting the stator, the camera may help show whether the sound source is concentrated near the frame or another structural region [9] [23] [24].

Effect of operating conditions

Healthy spectra are **condition-dependent**. Load, speed, voltage distortion, voltage imbalance, shaft speed, and housing sensitivity can all shift or amplify components [2] [3] [4] [5] [6]. Therefore, the correct methodological approach is not to ask "what is the normal frequency?" in the singular, but rather "what is the normal spectral baseline for this motor under this operating condition?" [5] [6].

Confidence level: High - There is strong convergence that healthy spectra are harmonic and condition-dependent, with clear physical explanations [2] [7] [8] [9].

Fault-Specific Frequency Signatures

Bearing-fault signatures

Bearing-related diagnosis appears in both acoustic-only and cross-modal studies. One acoustic-spectrum study compared healthy-motor and bearing-fault noise spectra using covariance-based subspace identification and cross-power spectrum methods [41]. Simultaneous vibration and acoustic-emission monitoring also found both signal types suitable for bearing fault detection, with **envelope spectra** discussed in terms of harmonic fault frequencies [36].

A broader pattern from the retrieved set is that bearing defects often produce **harmonic or envelope-based components** that differ from healthy spectra, but exact reproducible acoustic frequency tables were not provided in the extracted summaries [36] [41].

Rotor-bar and rotor-cage fault signatures

Broken rotor bars and squirrel-cage defects are repeatedly identified through **sidebands and harmonic changes** across current, vibration, and acoustic-related literature [12] [13] [14] [15] [18]. One experimental motor fault study specifically used a **4-pole, 50 Hz, 2.2 kW, 1420 rpm squirrel-cage induction motor with 28 rotor bars** and created the fault by drilling bars, although the retrieved snippet did not list the resulting acoustic frequencies [27].

Cross-modal evidence suggests that **current analysis is especially sensitive to broken rotor bars**, while acoustic analysis still detects faults under stationary and non-stationary conditions [34] [35]. For an ideathon project, this means broken-bar faults are promising if you want both a known spectral mechanism and cross-modal validation [18] [34].

Stator and winding-related faults

Acoustic diagnosis has also been applied to **loose stator coils**, **main/auxiliary winding shorts**, and other stator-related faults [23] [26] [42] [43]. A single-phase induction-motor acoustic study separated healthy condition from auxiliary/main winding shorts, broken rotor bar/squirrel-cage ring faults, and bearing fault using acoustic features and classification with reported recognition of 94% to 97% [26]. Another 2019 study diagnosed **bearing, stator, and rotor faults** in a single-phase induction motor using feature extraction with nearest-neighbour classification [43].

The limitation is interpretability: these studies support feasibility of acoustic discrimination, but the retrieved excerpts do not provide a full fault-frequency table [26] [43].

Eccentricity and mechanical misalignment

Static eccentricity, soft foot, misalignment, and foundation looseness were all reported as acoustically detectable or acoustically classifiable in the broader evidence [17] [20] [23] [44]. Fault studies identified sidebands and harmonics linked to **eccentricity** and related mechanical effects, with **load and mechanical resonance** changing observed components nonlinearly [5] [16] [17] [20].

This implies that eccentricity-related acoustic diagnosis may be especially sensitive to machine structure and mounting conditions, which is important if your ideathon prototype will be tested on different motor housings [5] [6].

Confidence level: Medium - The existence of fault-specific spectral behavior is well supported, but the extracted evidence does not consistently provide exact numeric acoustic frequencies for each fault class [27] [36] [41].

Acoustic Camera and Cross-Modal Fault Diagnosis

Acoustic camera and microphone-array methods

The retrieved studies support the use of **microphone arrays and acoustic-camera-style measurements** for localizing motor sound sources and separating fault-related sound from background noise [23] [24]. In direct industrial-style validation, acoustic-camera or microphone-array measurements were compared against vibration in **noisy environments**, and source localization helped isolate the fault-related sound field [23] [24] [25].

There is also evidence from rotating machinery more broadly that **MEMS microphone arrays** and **acoustic images** can be used for fault detection, with geometric features from those acoustic images feeding SVM classifiers [33].

Comparison with vibration and current monitoring

Cross-modal comparisons show complementary strengths:

- **Current** was reported as most sensitive to **broken rotor bars** [34].
- **Vibration** was reported as most sensitive to **bearing faults** [34].
- **Acoustic analysis** was still able to detect faults in both stationary and non-stationary conditions, and was described as having less noise/interference in the analyzed band in one comparison [34] [35].

Taken together, these findings suggest that an acoustic camera is particularly strong for **non-contact detection plus localization**, but the strongest technical pitch is a **hybrid system** where acoustic data are interpreted alongside vibration or current signatures [34] [36].

Feature extraction and classification pipelines

Acoustic-fault studies commonly use:

- **FFT**
- **STFT**
- **Wavelets**

- MFCC/cepstral features
- PCA/LDA
- Neural-network or nearest-neighbour classifiers [26] [30] [31] [37] [38] [39] [43].

But an important contradiction in the literature is that strong classification results do not automatically provide interpretable fault-frequency catalogs [30] [31] [32]. For your ideathon, this means a physics-linked spectral explanation could be a differentiator.

Confidence level: High - The evidence strongly supports non-contact localization value and complementary use with vibration/current monitoring [23] [24] [34].

Experimental Conditions, Limitations, and Best Paper Selection

Reproducibility requirements

For a paper to be maximally useful to your project, it should report:

- Machine type and rating
- Fault type introduced
- Sampling rate
- Record length
- Operating speed
- Load condition
- Number and placement of microphones
- Spectral resolution or analysis method [1] [27].

Among the retrieved evidence, the strongest match on reproducibility is the induction-motor experiment conducted in a reverberant noisy room with five microphones, 44.1 kHz sampling, and about 30 s steady-state recordings [1]. That level of setup detail is directly useful for project design [1].

Noise and machine-structure limitations

The literature also shows important limits. Background noise, reverberation, enclosure effects, housing sensitivity, shaft speed, and machine-specific structure can all distort the measured acoustic signature [6] [44] [45]. This is one reason why universal fault-frequency tables are hard to standardize across machines [6].

Most relevant paper for the project

Based on the supplied evidence, the most related paper is the one described in [1], because it best fits your actual project requirements:

- **Acoustic measurement rather than only current/vibration** [1]
- **Healthy and faulty motors compared** [1]
- **Multiple relevant faults**: broken rotor bars, bearing defects, eccentricities [1]
- **Realistic noisy environment** [1]
- **Multi-microphone setup**, which is the closest retrieved evidence to an acoustic-camera workflow [1]

If you need to justify why this is not fully sufficient alone, the answer is simple: the retrieved evidence says no single paper clearly combines **all healthy baseline frequencies**, **all major fault frequencies**, and **detailed acoustic-camera settings** in one place [1] [27] [28] [29].

Confidence level: High - The recommendation follows directly from the evidence description and from the explicit gap statement in the retrieved set [1] [29].

Practical Recommendation for Your Ideathon

A strong evidence-based project plan would be:

1. **Use [1] as your anchor paper** for the experimental setup and justification of microphone-based motor fault diagnosis in realistic conditions [1].
2. **Define healthy baseline spectra** using literature on electromagnetic noise, line-frequency harmonics, and resonance [2] [3] [7] [8] [9].
3. **Interpret faults through sidebands and harmonic additions** reported for rotor-bar, stator, eccentricity, and mechanical faults [15] [16] [17] [18] [19] [20].
4. **If possible, validate with vibration or current** because cross-modal evidence shows they help confirm fault type [34] [36].
5. **Emphasize low-frequency content as well as localized sound maps**, since sub-100 Hz information may be important [40] [41].

An inferred but practical synthesis is that your ideathon can stand out by building the thing this literature lacks: a **fault-frequency atlas tied to acoustic localization**. **Inferred:** Taken together, the studies suggest that the gap is not whether acoustic diagnosis works, but whether it can be made **interpretable, reproducible, and condition-aware** in one unified workflow [1] [32] [34].

Confidence level: Medium - This final integration is a synthesis across sources rather than a direct statement from one paper [1] [32] [34].

Conclusion

The retrieved evidence supports a clear answer: the **closest paper to your need** is the **realistic multi-microphone induction-motor study** summarized in [1]. It is the best match because it combines **healthy versus faulty operation**, **multiple fault classes**, and a **microphone-based experimental setup** under noisy conditions [1].

At the same time, the evidence also shows that **normal machine frequency is condition-dependent**, not a single value, and that fault diagnosis depends on identifying **harmonics and sidebands** relative to that baseline [2] [3] [5]. So the best research-backed approach for your project is to combine:

- one **setup-focused acoustic paper** [1],
- several **healthy-spectrum mechanism papers** [7] [8] [9],
- and several **fault-signature papers** [15] [16] [18] [19].

Overall confidence level: High - The recommendation is directly grounded in the retrieved evidence, and the main uncertainty is not about relevance but about the literature gap in unified frequency tables [1] [29].

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